Exploring Word Clouds: Taking a Deeper Look at How They Interact with Middle School Students' Data and Literary Meaning-Making Processes

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Word clouds have a complicated role in the digital humanities ecosystem—they're a common "first" visualization for digital humanists, especially in postsecondary education contexts as they're built into tools like Voyant (Sinclair, S. / Rockwell, G. 2016). However, they're inconsistently used and widely criticized (Felix et al. 2017; Hicke et al. 2022). This project pivots from looking at how researchers do or should use word clouds to understand how K-12 students and teachers engage with word clouds, taking a descriptivist, rather than prescriptivist, approach. The research centers on two key questions: a) how do students make sense of word clouds from a data literacy perspective, and b) how do word clouds interact with their literary meaning-making processes?

The context for this study is a three-year initiative conducted in collaboration with a team of university researchers and a California public secondary school district. This project investigates how digital humanities techniques, such as numerical and text-analytics based visualizations, can support multidimensional literacies—integrating literary and data literacy—for middle school students. Grounded in teacher curricula and district English Language Arts (ELA) goals, we designed visualizations, including word clouds, bar charts, and line graphs, to support the texts already being taught in these classrooms. For the past two years, we have been working iteratively with ELA teachers on data visualizations and their potential contributions to student learning.

This initial research draws on a set of qualitative interviews with middle-school students that ask them to read a word cloud, read the poem used to generate that word cloud, and then compare the two textual representations. Students were interviewed at the start of the academic year, before regular exposure to the intervention's data-literacy lessons. How do these students make sense of word clouds? What do they notice, what do they draw on to make meaning, and what do they critique? We use qualitative coding to explore these questions in the context of these interviews. We find that, in line with the encoding practices discussed in Felix et al. (2017), students draw on word color, size, and position to make meaning of word clouds. We also find examples of students using word clouds to make predictions, to add connotative meaning, and to draw connections. They occasionally criticized word clouds for their layout and other design choices, and in comparing the poem and word cloud, several changed their interpretations of the word cloud.

By understanding how students make sense of word clouds without specific training in doing so, we hope to surface the affordances and risks of this form of visualization in non-specialist contexts. This preliminary work is ultimately in support of producing recommendations for both the design of, and pedagogical engagement with, word clouds to support literary-critical meaning making practices and developing student data literacy.

Bibliography

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